

HW4

2025-11-13

3230106230 刘仁钦

1. Bit rate per station: $R = \frac{1000 \text{ bit}}{100 \text{ sec}} = 10 \text{ bps}$

Maximum utilization for ALOHA: $\frac{1}{2e}$

Maximum throughput: $56 \text{ kbps} \times \frac{1}{2e} \approx 10300.62 \text{ bps}$

$N \times R \leq 10.3 \text{ kbps} \Rightarrow N \leq 1030$

Maximum value of N is 1030

2. The Manchester encoding is as Figure 1. Using positive pulse to represent 1 and negative pulse to represent 0.

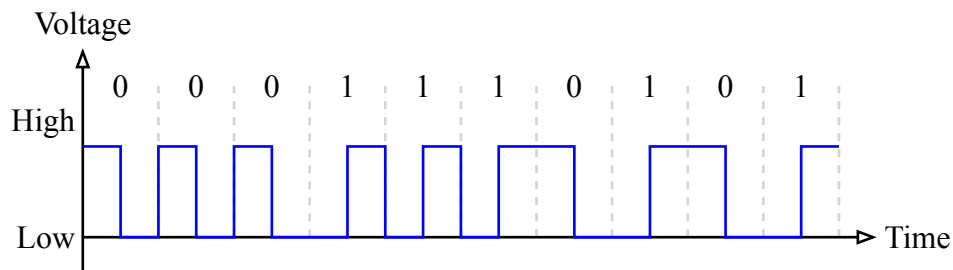


Figure 1: Manchester encoding for 0, 0, 0, 1, 1, 1, 0, 1, 0, 1

3. Round-trip propagation delay: $T_r = 2 \times \frac{d}{v} = 2 \times \frac{1000\text{m}}{2 \times 10^8 \text{ m/s}} = 10^{-5} \text{ s}$

Minimum frame transmission time: $T_t = \frac{L}{R} \geq T_r$

Therefore $L \geq R \times T_r = 10^9 \text{ bps} \times 10^{-5} \text{ sec} = 10000 \text{ bits}$

$\frac{L}{8} = 1250 \text{ bytes}$

Thus, the minimum frame size is 1250 bytes

4. The standard 10-Mbps Ethernet uses Manchester encoding. In Manchester encoding, each bit is represented by two signal changes (one high-to-low or low-to-high transition). Therefore, the baud rate (symbol rate) is twice the data rate.

For 10-Mbps Ethernet:

Data rate = 10 Mbps

Baud rate = $2 \times 10 \text{ Mbps} = 20 \text{ Mbaud}$

Thus, the baud rate of the standard 10-Mbps Ethernet is 20 Mbaud

5. C

6. B

7. B

- 8. C
- 9. B
- 10. C